G H Raisoni International Skill Tech University, Pune

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End Semester Examinations- Winter 2025

School of Engineering and Technology

F.Y. B.Tech. Semester - I (All Branches)

Course Name: Foundation of Data Analytics **Course Code: 25UESL1206**

Max Marks: 60 **Duration: 150 minutes**

Instructions:

1. Write the solution in detail.
2. Clearly show all steps of your work, including necessary derivations and calculations.
3. Read each question carefully before attempting.
4. Students must label their answers with the corresponding question number, such as Q.No. 1A, Q. No. 2C, etc.
5. Each question is of 12 marks, from every question solve any **TWO from A, B & C**. Question **D** is **compulsory**.

Q. No.	Sub Q.No.	Question	Marks	CO	BL
1		Solve any **TWO from Question A, B & C**. Question **D is compulsory.**			
	A	What is Range, Variance, Standard Deviation and Quartiles? Explain With example.	5	2	2
	B	Distinguish between Quantitative and Qualitative analytics with suitable examples.	5	1	2
	C	Explain the different types of data analytics with real-world examples.	5	1	2
	D	Find Variance, Standard Deviation and Range of data-{2,3,4,5,6,7,8,10}	2	2	3
2		Solve any **TWO from Question A, B & C**. Question **D is compulsory.**			
	A	What are the advantages of using CSV and Excel files for data storage and import?	5	2	2
	B	Describe the process of importing data from a database table into Python with an example.	5	2	3
	C	Why the following text functions are used in Excel: LEFT, RIGHT, LEN, MID, and FIND.	5	2	2
	D	What will be output of following python code? Left("Good Afternoon",6)	2	2	3
3		Solve any **TWO from Question A, B & C**. Question **D is compulsory.**			
	A	What is Margin of Error?	5	3	1
	B	Explain the Central Limit Theorem.	5	3	2
	C	Explain the difference between Simple and Multiple Linear Regression.	5	3	2
	D	State relation among SST (Sum of Squares Total), SSR (Sum of Squares	2	3	2

Regression), and SSE (Sum of Squares Error).

4		Solve any **TWO from Question A, B & C**. Question **D is compulsory.**	5	4	2
	A	How does Pivot Tables help in efficient data summarization and comparison?	5	4	2
	B	Describe the steps to create a frequency distribution using a Pivot Table.	5	4	3
	C	What is Frequency Analysis? Explain how it can be performed using a Pivot Table.	2	4	3
	D	Give example of a real-world scenario for ranking analysis in Pivot Tables are useful.			
5		Solve any **TWO from Question A, B & C**. Question **D is compulsory.**			
	A	Discuss how tabular data can be linked to charts and dashboards for dynamic reporting.	5	4	3
	B	How can charts help in analyzing trends and comparisons in data? Illustrate with an example.	5	4	3
	C	What is conditional formatting? Explain how it can be used to highlight high/low and above/below average values in a report.	5	4	3
	D	List steps to publish Power BI Report.	2	4	3

- End of the Question Paper -